

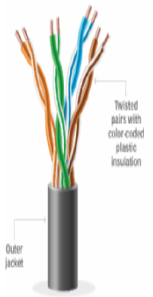
COMPUTER NETWORKS

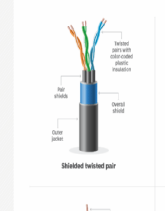
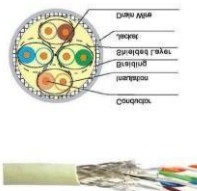
Ex.No.2	Study of Network cables. 1. Understand different types of Network cables. 2. Make a cross-wired cable and straight through cable using a clamping/crimping tool.
Date:	

a) **Understand different types of network cable.**

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	<u>Advantages</u> • Cheaper in cost • Easy to install as they have a smaller overall diameter.	10Base-T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps	<u>Disadvantages</u> • More prone to (EMI) Electromagnetic interference and noise	Fast Ethernet, Gigabit Ethernet	

STP	Category 6, 6a	10Gbps	Advantages - Shielded. - Faster than UTP. - Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps	Disadvantages - Expensive - Greater installation effort	Gigabit Ethernet, 10G Ethernet (100m)	

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	- High bandwidth - Immune to interference - Low loss bandwidth - Versatile Disadvantages - Limited distance - Cost - Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	Advantages - High speed - High bandwidth - High security - Long distance Disadvantages - Expensive - Requires	Maximum distance of fibre optics cable is around 100meters	

			skilled installers		
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(b) Make Your Own Ethernet Cross-Over Cable/ Straight cable

Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

Ethernet Cable (CAT5e/CAT5)

- CAT5e is recommended because it supports Gigabit Ethernet (1000 Mbps).
- CAT5 also works but supports lower speeds and shorter distances.

Crimping Tool

- A networking tool used to:
 - Strip the outer insulation of the cable.
 - Arrange and trim the wires.
 - Crimp (press) the RJ45 connector onto the cable.

RJ45 Connectors (2)

- Plastic connectors fitted at both ends of the Ethernet cable.
- Used to connect the cable to computers, switches, routers, etc.

Plug Shields (Optional)

- Protective covers placed over the RJ45 connectors.
- Prevent damage to the locking clip and reduce strain on the cable.

How to Make an Ethernet Cable (Straight-Through / Cross-Over)

Tools & Parts



CAT5e / CAT6 Cable



RJ45 Connector



Crimping Tool



Cable Stripper (Optional)

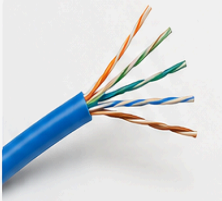
1 Cut the Cable

Cut the Ethernet cable to the desired length.



2 Strip the Outer Jacket

Use the cable stripper to remove about 2-3 cm (1 inch) of the outer jacket.



3 Untwist and Straighten the Wires

Untwist the wires and straighten them.



4 Arrange the Wires (Choose a Standard)

Straight-Through Cable: Use the same standard on both ends (T568B or T568A).

T568B Standard							
1	2	3	4	5	6	7	8
White/Orange	Orange	White/Green	Blue				
White/Blue	Green	White/Brown	Brown				



5 Trim the Wires

Trim the wires evenly so that they are all the same length.



6 Insert into RJ45 Connector

Insert the wires into the RJ45 connector. Make sure each wire goes all the way to the end and the order is correct.



7 Crimp the Connector

Place the connector into the crimping tool and squeeze firmly to crimp it onto the cable.



8 Repeat on the Other End

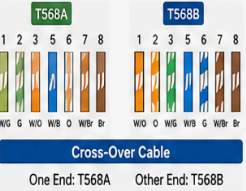
Repeat the same steps on the other end of the cable.

- For Straight-Through: Use the same standard as the first end.
- For Cross-Over: Use T568A on one end and T568B on the other.

T568A				T568B			
1	2	3	4	1	2	3	4
WG	G	WB	Br	WG	G	WB	Br
WB	O	WBr	Br	WB	O	WBr	Br

Cross-Over Cable

One End: T568A Other End: T568B



9 Test the Cable

Use a cable tester to check if all 8 wires are connected properly.



10 Finished

Your Ethernet cable is ready to use!



Straight-Through Cable (Same Standard on Both Ends)



Computer — Switch / Hub — Router

Cross-Over Cable (T568A to T568B)



Computer — Computer Switch — Switch

Tips

- Make sure the wire order is correct.
- Push wires all the way to the end of the RJ45 connector.
- Crimp firmly for a good connection.

RESULT: Hence different types of network cables and their uses and differences has been studied.